BiSI Thin-Film XRD Analysis and Bayesian Optimization

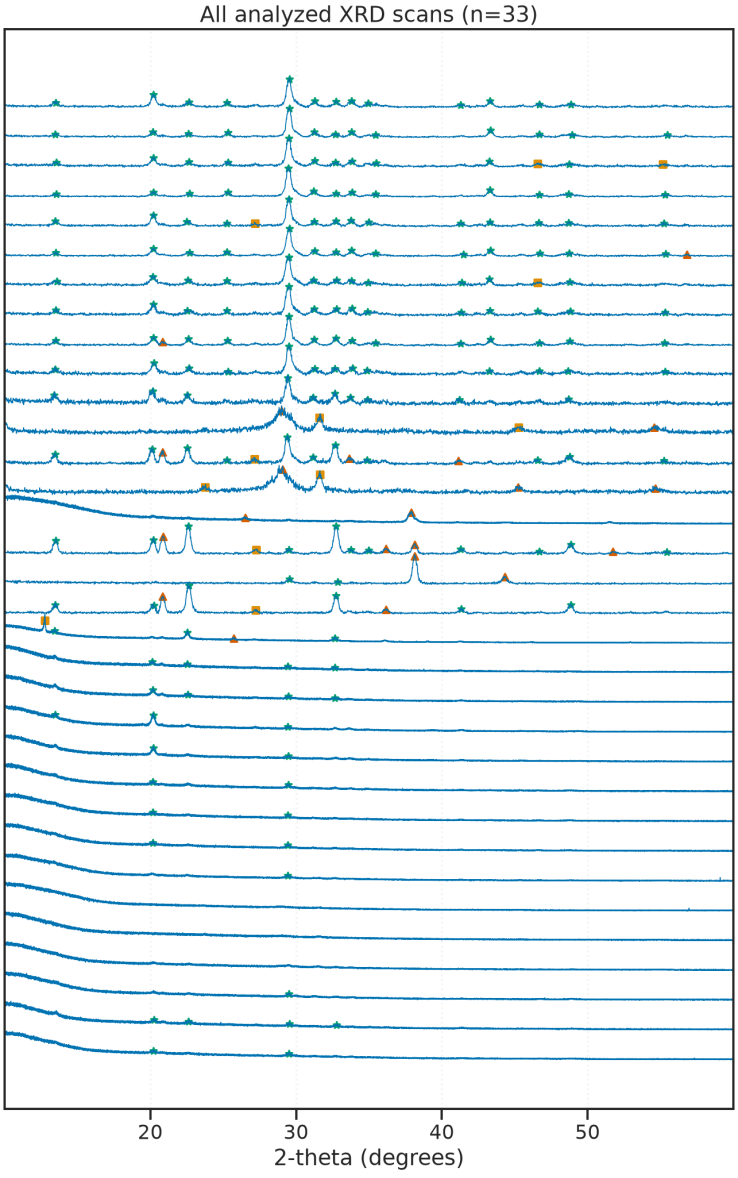
NOMAD-driven extraction, phase screening, and exploratory synthesis planning

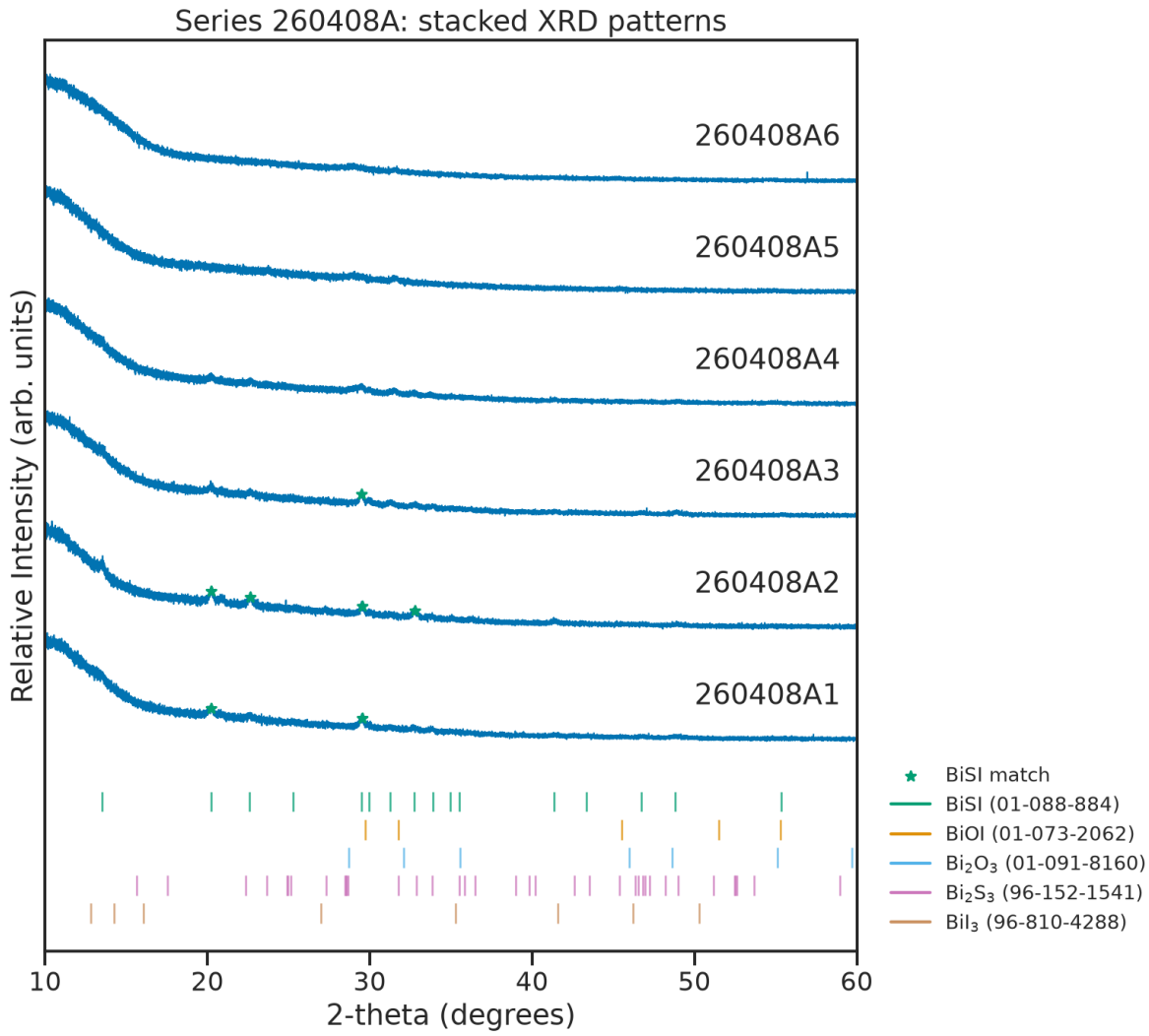
1. Dataset and method

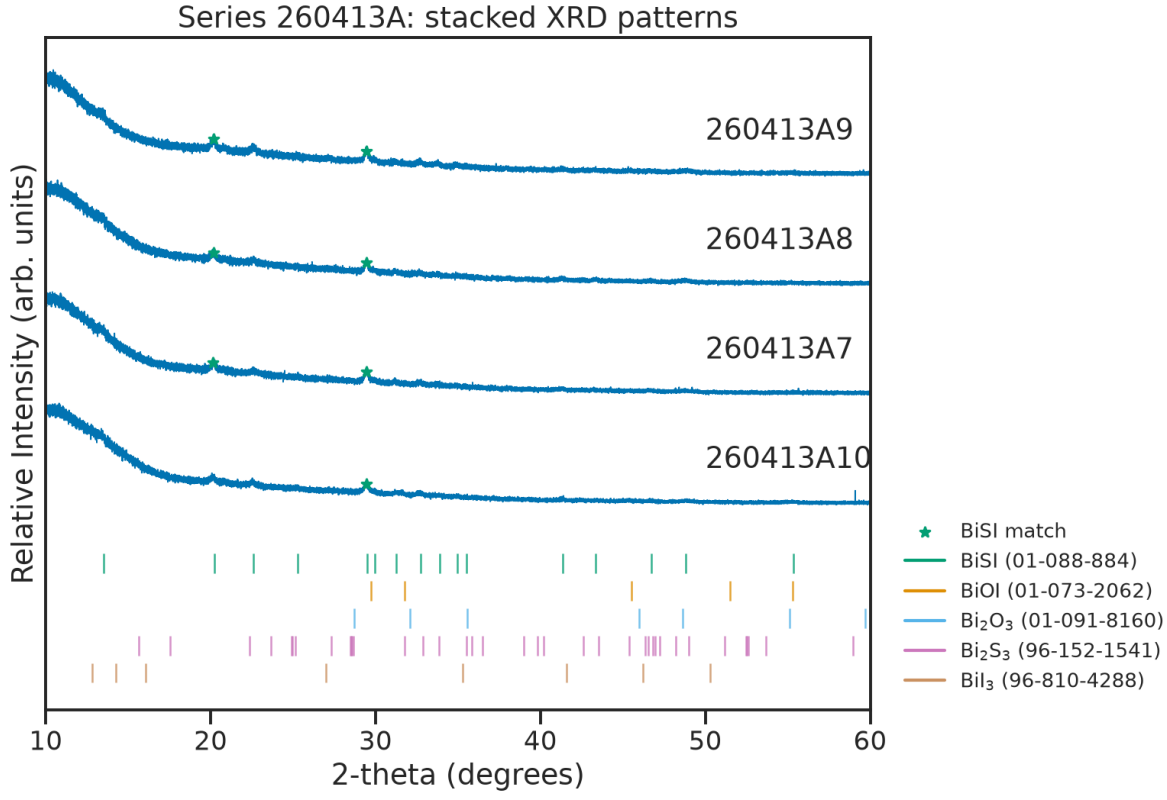
The BiSI NOMAD extraction contains 41 thin-film records; 39 have matched main-layer precursor chemistry and 41 have seed-solution chemistry. XRD analysis produced 33 scan metric rows, with 25 complete observations for the optimizer. The target phase is BiSI; the configured candidate phases are BiOI, Bi2O3, Bi2S3, and BiI3. Composition metrics are calculated from NOMAD solution solute formulas and amounts: Bi molar concentration and atomic S:Bi and I:Bi ratios. Crystallinity is the inverse FWHM of the strongest matched target peak. An optional Le Bail proxy converged for 13 scans; 20 scans lacked an exact matching GIXRD archive-array record.

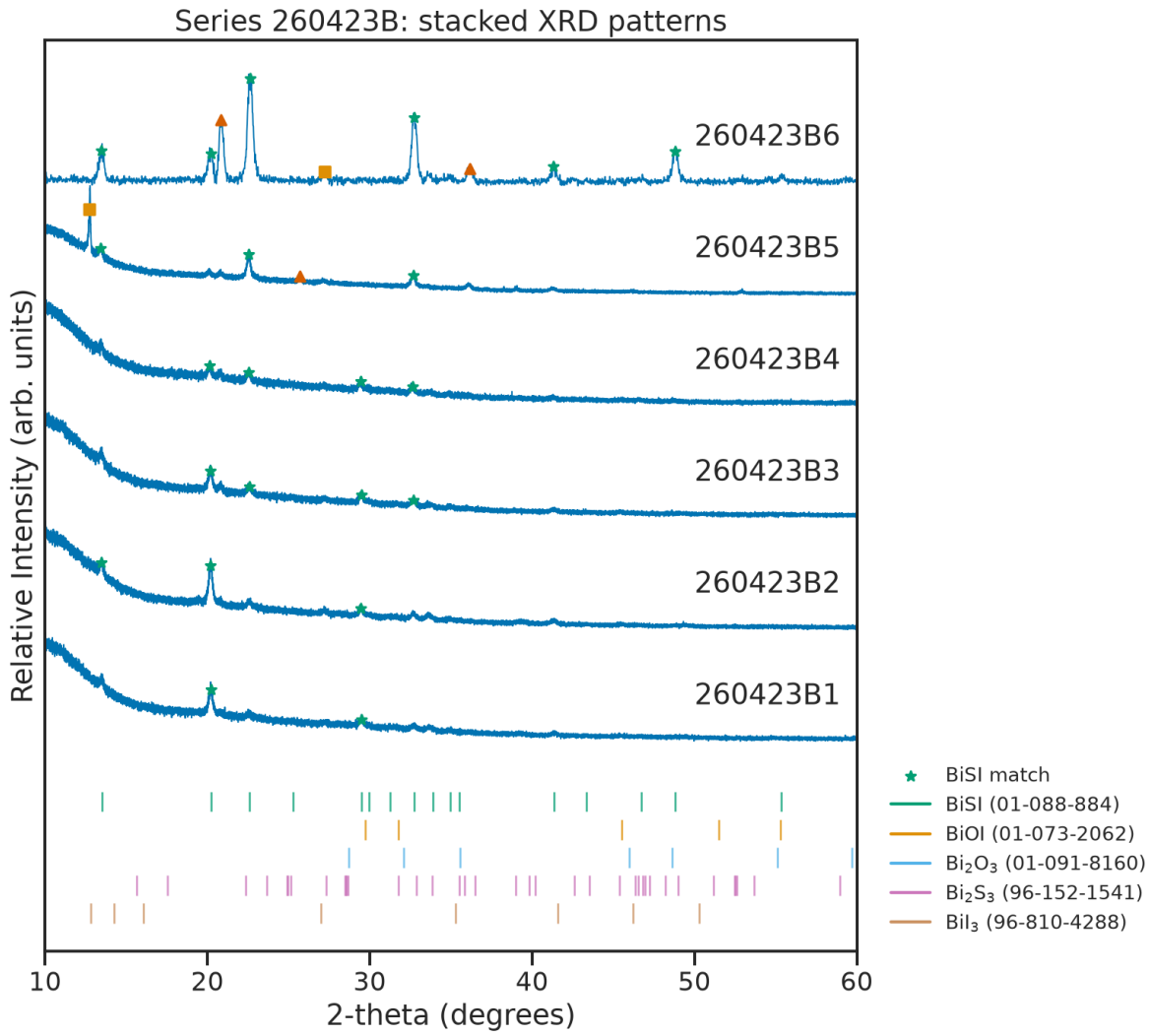
2. XRD phase screening

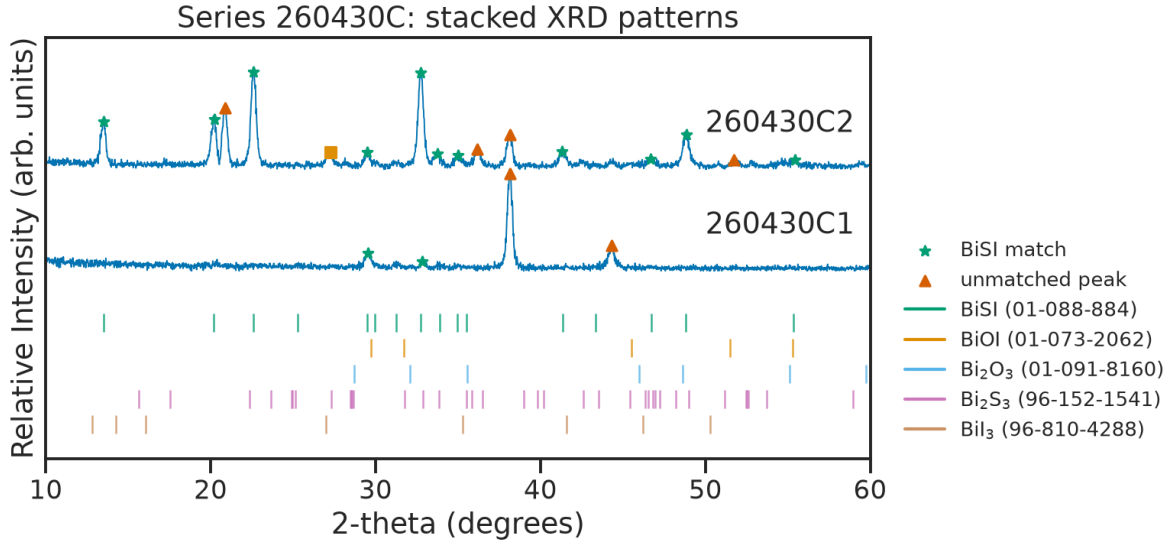
Stacked BiSI XRD patterns per series with reference bars for the target and candidate phases. Markers flag target (star), impurity (square), and unmatched (triangle) peaks.

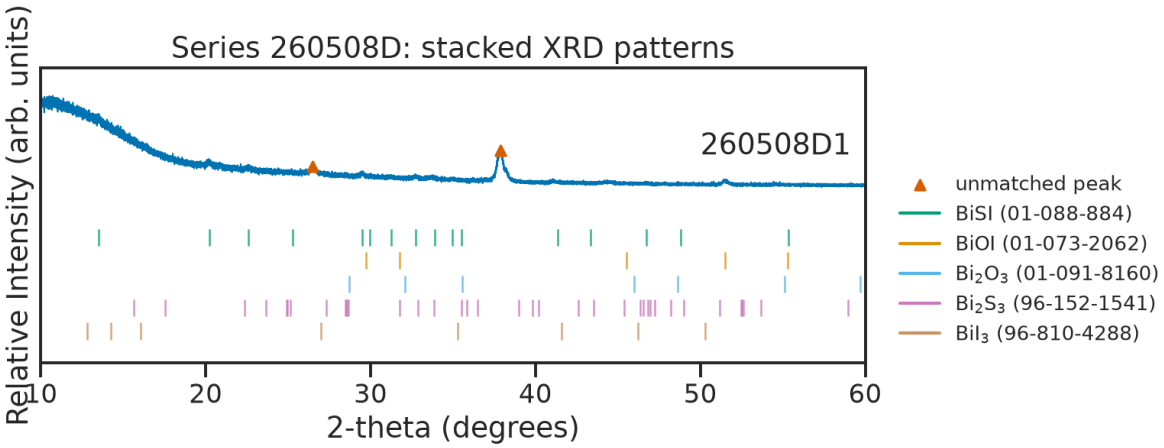


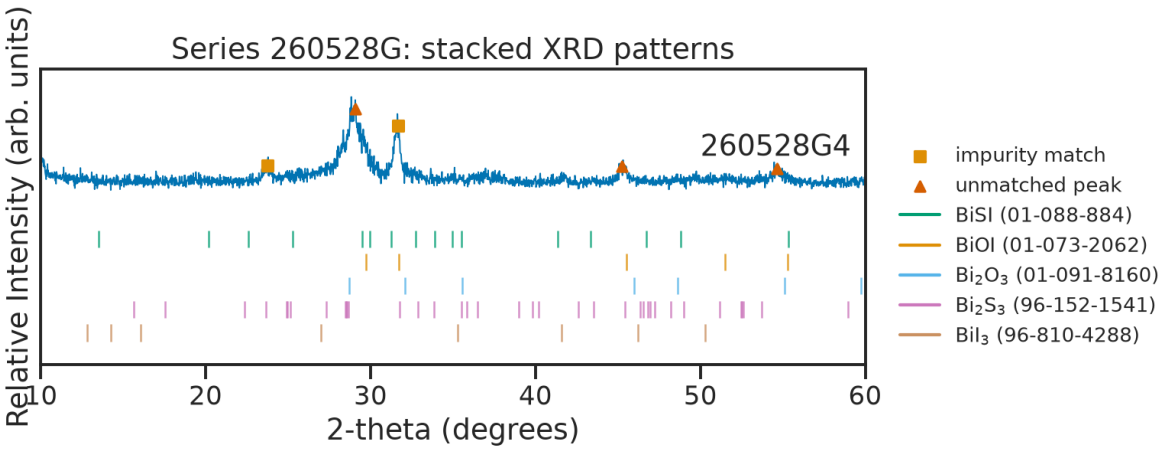

Series 260408A: stacked XRD patterns
260408A6
260408A5
260408A4
260408A3
260408A2
260408A1
Relative Intensity (arb. units)
10
20
30
40
50
60
2-theta (degrees)
BiSI match
BiSI (01-088-884)
BiOI (01-073-2062)
Bi2O3 (01-091-8160)
Bi2S3 (96-152-1541)
BiI3 (96-810-4288)

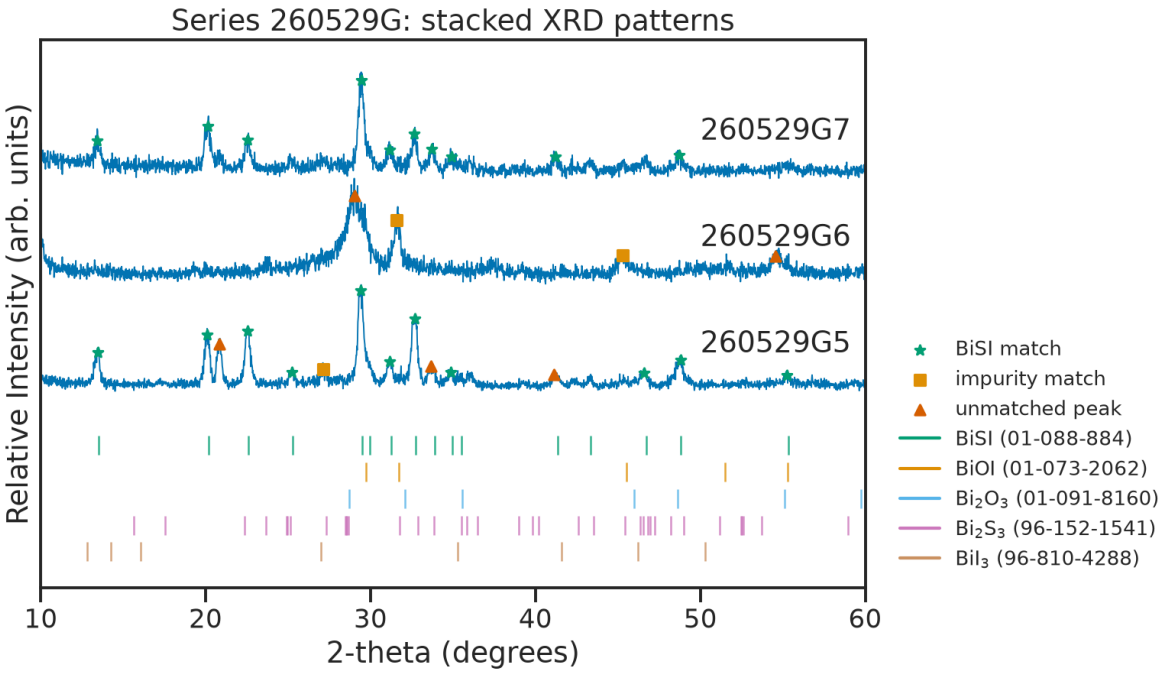

Series 260413A: stacked XRD patterns
260413A9
260413A8
260413A7
260413A10
Relative Intensity (arb. units)
10
20
30
40
50
60
2-theta (degrees)
BiSI match
BiSI (01-088-884)
BiOI (01-073-2062)
Bi2O3 (01-091-8160)
Bi2S3 (96-152-1541)
BiI3 (96-810-4288)

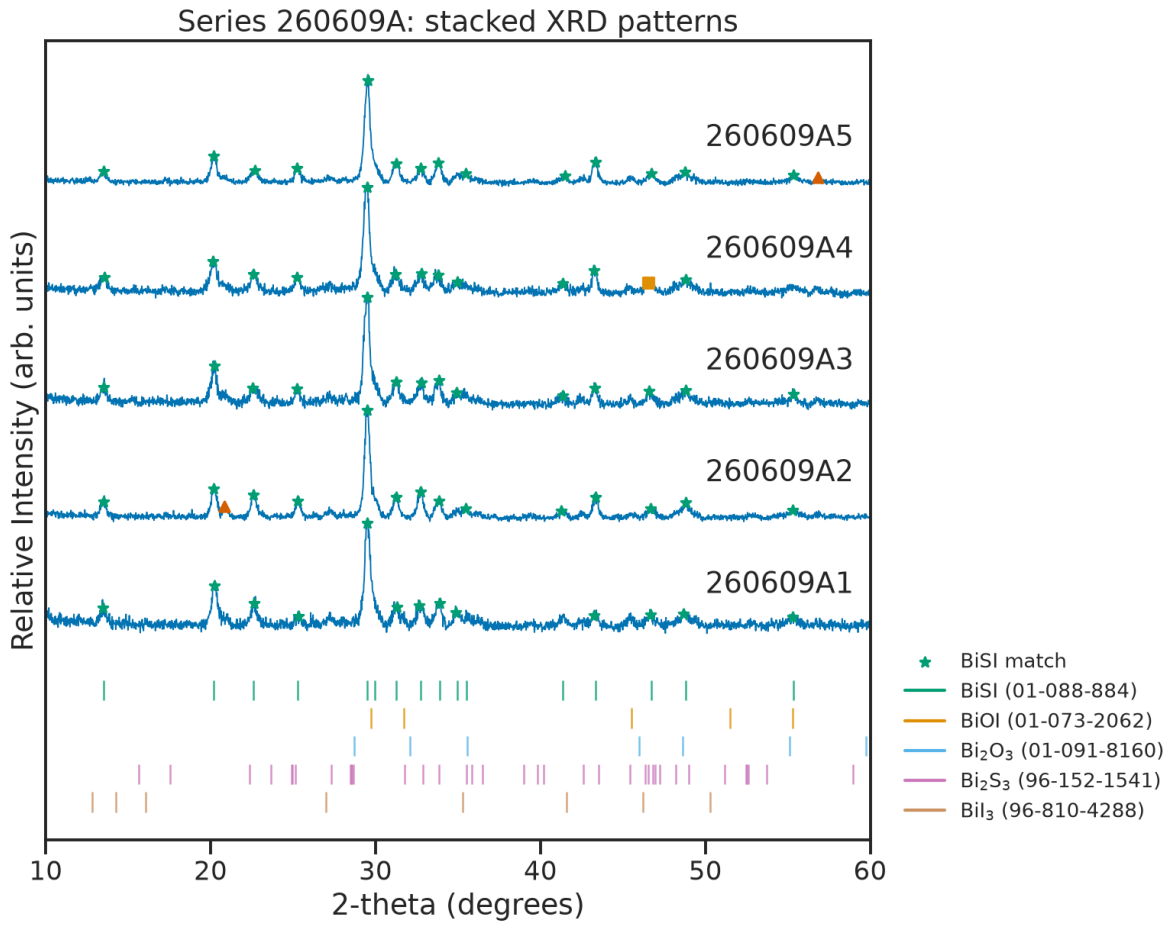

Series 260423B: stacked XRD patterns
260423B6
260423B5
260423B4
260423B3
260423B2
260423B1
Relative Intensity (arb. units)
10
20
30
40
50
60
2-theta (degrees)
BiSI match
BiSI (01-088-884)
BiOI (01-073-2062)
Bi2O3 (01-091-8160)
Bi2S3 (96-152-1541)
BiI3 (96-810-4288)

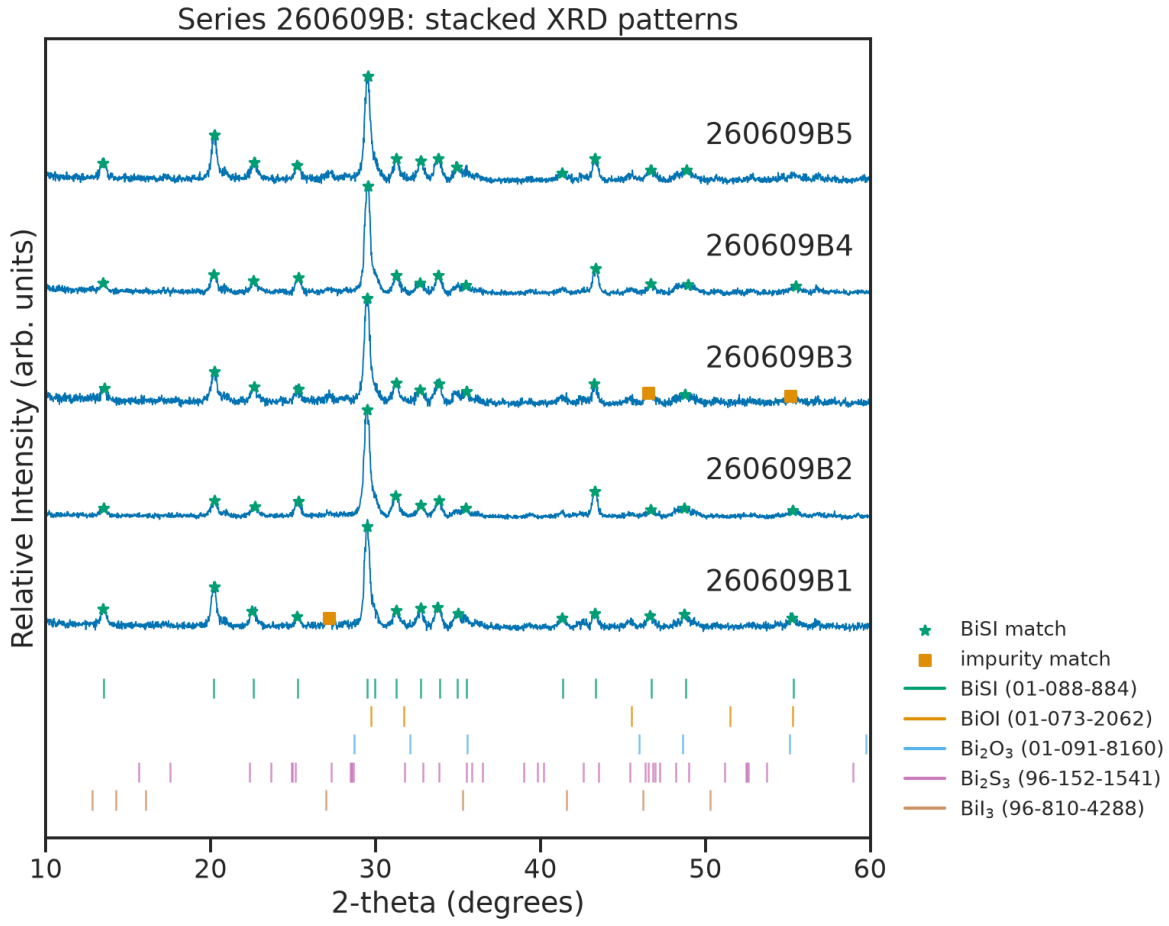

Series 260430C: stacked XRD patterns
260430C2
260430C1
Relative Intensity (arb. units)
10
20
30
40
50
60
2-theta (degrees)
BiSI match
unmatched peak
BiSI (01-088-884)
BiOI (01-073-2062)
Bi2O3 (01-091-8160)
Bi2S3 (96-152-1541)
BiI3 (96-810-4288)


Series 260508D: stacked XRD patterns
Relative Intensity (arb. units)
260508D1
10
20
30
40
50
60
2-theta (degrees)
unmatched peak
BiSI (01-088-884)
BiOI (01-073-2062)
Bi2O3 (01-091-8160)
Bi2S3 (96-152-1541)
BiI3 (96-810-4288)


Series 260528G: stacked XRD patterns
Relative Intensity (arb. units)
260528G4
10
20
30
40
50
60
2-theta (degrees)
impurity match
unmatched peak
BiSI (01-088-884)
BiOI (01-073-2062)
Bi2O3 (01-091-8160)
Bi2S3 (96-152-1541)
BiI3 (96-810-4288)

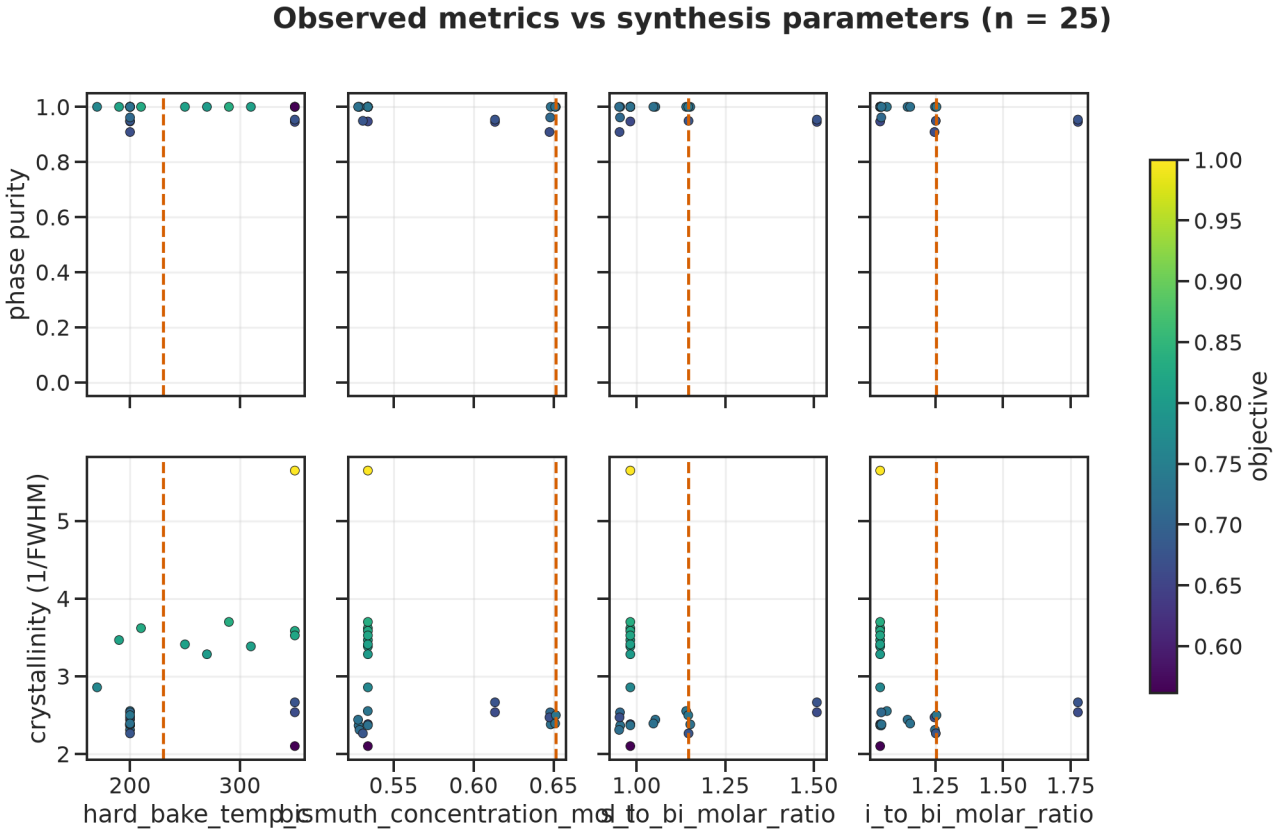

Series 260529G: stacked XRD patterns
Relative Intensity (arb. units)
260529G7
260529G6
260529G5
10
20
30
40
50
60
2-theta (degrees)
BiSI match
impurity match
unmatched peak
BiSI (01-088-884)
BiOI (01-073-2062)
Bi2O3 (01-091-8160)
Bi2S3 (96-152-1541)
BiI3 (96-810-4288)


Series 260609A: stacked XRD patterns
260609A5
260609A4
260609A3
260609A2
260609A1
Relative Intensity (arb. units)
10
20
30
40
50
60
2-theta (degrees)
BiSI match
BiSI (01-088-884)
BiOI (01-073-2062)
Bi2O3 (01-091-8160)
Bi2S3 (96-152-1541)
BiI3 (96-810-4288)


Series 260609B: stacked XRD patterns
260609B5
260609B4
260609B3
260609B2
260609B1
Relative Intensity (arb. units)
10
20
30
40
50
60
2-theta (degrees)
BiSI match
impurity match
BiSI (01-088-884)
BiOI (01-073-2062)
Bi2O3 (01-091-8160)
Bi2S3 (96-152-1541)
BiI3 (96-810-4288)

3. Observed results

For the 25 complete observations, mean phase purity is 0.987 and mean crystallinity is 2.909. The highest screening-adjusted score is 1.000 for 260423B2 (phase purity 1.000, crystallinity 5.647, unmatched/overlap area proxy A=0.000). The score is an exploratory ranking metric, not a phase-fraction determination. It uses (0.7 × phase purity + 0.3 × normalized crystallinity) × (1 – 0.25 × A).



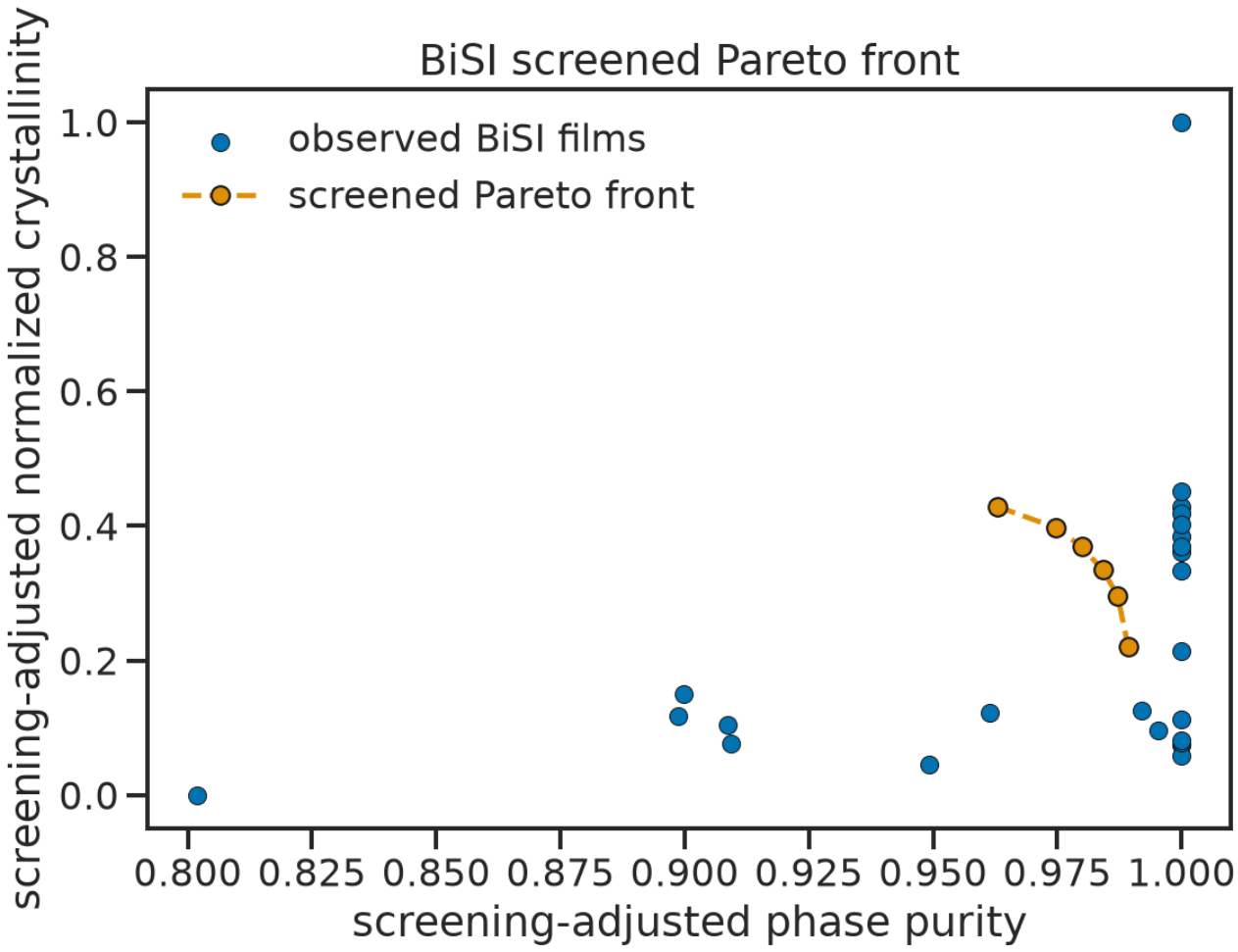
4. Single-objective candidate recipes

The Gaussian-process candidates use measured levels for the composition variables and a hard-bake grid spanning the observed conditions. Predictions are model estimates, not validated performance or ready-to-run recipes:

#	Hard bake °C	Bi concentration mol/L	S:Bi molar ratio	I:Bi molar ratio	Predicted score	GP std
1	230	0.6515	1.146	1.254	0.767	0.042
2	230	0.5338	1.140	1.157	0.767	0.042
3	230	0.5278	0.953	1.147	0.767	0.042
4	230	0.6482	0.952	1.052	0.767	0.042
5	230	0.6510	1.146	1.051	0.767	0.042

5. Multi-objective candidates

Ax optimizes the screened phase-purity and crystallinity objectives separately. All input values are selected from observed levels; proposals are exploratory and should be checked against process feasibility.



#	Hard bake °C	Bi concentration mol/L	S:Bi molar ratio	I:Bi molar ratio
1	350	0.5339	1.509	1.047
2	350	0.6134	1.151	1.047
3	350	0.5278	0.951	1.776
4	350	0.5278	0.951	1.254
5	170	0.6134	1.151	1.776

Interpretation limits

Peak classification depends on the selected RTF cards, intensity threshold, peak detector, and matching tolerance. The unmatched/overlap area is a screening proxy, not a quantitative phase fraction. The Le Bail fits use a Bragg–Brentano profile model as a proxy for GIXRD and are qualitative only; fitted intensities do not establish validated phase fractions. Verify phase references and process feasibility before choosing follow-up experiments.